

CS 241: Introduction to Cryptography

Course Code: CS 241

Course Name: Introduction to Cryptography

L-T-P-C: 3-0-0-3

Category:

Prerequisites: None

Approval: 5th Senate

Course Contents

- **Overview of Cryptography**
- Introduction, Information Security and Cryptography, Background on Functions, Basic Terminology and Concepts, Symmetric Key Encryption, Digital Signatures, Authentication and Identification, Public Key Cryptography, Hash Functions, Protocols and Mechanisms, Classes of Attacks and Security Models
- **Classical Cryptography**
 - Introduction to Some Simple Cryptosystems, The Shift Cipher, The Substitution Cipher, The Affine Cipher, The Vigenere Cipher, The Hill Cipher, The Permutation Cipher, Stream Ciphers
 - Cryptanalysis, Cryptanalysis of the Affine Cipher, Cryptanalysis of the Substitution Cipher, Cryptanalysis of the Vigenere Cipher, A Known Plaintext Attack on the Hill Cipher
 - Public Key Cryptography, Introduction to public key cryptography, Number theory, Algebra, RSA, DHP and Discrete Log assumptions, Diffie Hellman key exchange, RSA public key system, ElGamal encryption, Pseudo-random bit generators
 - Digital Signatures, Digital signatures: definitions and applications, How to sign using RSA, Overview of signatures based on discrete-log
- **Basic Symmetric Key Encryption**
 - One time pad and stream ciphers, Shannons Theory, Block Ciphers, Case studies: Feistel networks, DES, 3DES, and AES, Basic modes of operation: CBC and counter mode
 - Attacks on Block Ciphers, exhaustive search, time-space tradeoffs, differential & linear cryptanalysis, meet in the middle, side channels
 - Message Integrity, Message integrity: definition and applications, Collision resistant hashing, Merkle-Damgard and Davies-Meyer. MACs from collision resistance, Case studies: SHA and HMAC

Text Books:

1. Abhijit Das and C. E. VeniMadhavan, **Public-Key Cryptography: Theory and Practice**, Pearson Education
2. Ivan Niven, Herbert S. Zuckerman and Hugh L. Montgomery, **An Introduction to the Theory of Numbers**, Wiley-India
3. I. N. Herstein, **Topics in Algebra**, 2nd Edition, Wiley India.